

Mingling Yang

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EDUCATION

University of Southern California, Los Angeles, CA

Sep 2026 - Present

Ph.D in Epidemiology in Environmental Health Track; Keck School of Medicine

- Relevant Courses: Data Analysis, Methods in Epidemiology I&II, Statistical Methods for Epidemiological Studies I&II

Johns Hopkins University, Baltimore, MD

Sep 2021 - May 2023

M.S.E in Applied Mathematics and Statistics; Whiting School of Engineering

- Relevant Courses: Mathematical Biology, Computational Molecular Medicine, Bayesian Statistics, Mathematical Image Analysis, Time Series Analysis, Probability Theory, Data Science, Machine Learning, Numerical Analysis
- Certifications & Training: Biostatistics in Public Health Specialization and Epidemiology in Public Health Practice Specialization (Johns Hopkins University); Causal Inference (Columbia University); Study Designs in Epidemiology (Imperial College London)

University of South Carolina, Columbia, SC

Jan 2017 - Dec 2020

B.S. in Statistics, GPA: 3.86/4.00 (rank 1%); McCausland College of Arts and Sciences

Minor in Risk Management and Insurance, GPA: 4.00/4.00 (rank 1%)

- Relevant Courses: Statistical Methods I&II&III, Statistical Inference, Calculus I&II&III, Linear Algebra, Statistics Inference, Mathematical Statistics, Algorithm Design in Java, SAS Language Programming, Stochastic Processes, Multivariate Statistics and Data Mining, Statistical Computing

HONORS AND AWARDS

- Multi-Year Funding Offer (\$697,000 total, 5-year, merit-based), University of Southern California 2026 - 2031
- Departmental Tuition Fellowship (\$8,551.5, merit-based), Johns Hopkins University 2021 - 2023
- USC Nonresident Scholarship (\$10,548/yr, merit-based, Top 1%), University of South Carolina 2019 - 2020
- USC Academic Achievement Scholarship (\$2,500, merit-based), University of South Carolina 2019 - 2020
- Dean's Honor List (Top 1%), University of South Carolina 2017 - 2020
- President's Honor List (Top 1%), University of South Carolina 2017 - 2020

PUBLICATIONS AND CONFERENCES

Peer Reviewed Journal Articles:

1. Checkley W, **Mingling Yang**, Robertson NM, Sharma AK, Chandyo RK, Shrestha L, Das SK, Kirenga B, Alupo P, Gianella G, Siddharthan T, Pollard SL, Quaderi S, Rykiel N, Flores-Flores O, Hurst JR, Wise RA; Global Excellence in COPD outcomes (GECO) study investigators. Population-based Screening for Chronic Obstructive Pulmonary Disease Using the St. George's Respiratory Questionnaire in Resource-limited Settings. *American Journal of Respiratory and Critical Care Medicine*. 2025 May; 211(5):779-788. doi: 10.1164/rccm.202409-1862OC. PMID: 40068167; PMCID: PMC12091018.(Featured in editorial)
2. Reuland CJ, Nicolaou L, Black MM, **Mingling Yang**, McCord R, Alvarado M, Arana F, Sanchez PS, Williams KN, Hossen S, Chiang M, Hartinger SM, Checkley W. Effects of Cooking with Liquefied Petroleum Gas or Biomass Fuels on Neurodevelopmental Outcomes in Preschool-Age Children Living in Puno, Peru. *Environmental Health Perspectives*. 2025 Jun;133(6):67004. doi: 10.1289/EHP15500. Epub 2025 Jun 5. PMID: 40354572; PMCID: PMC12142655.
3. Robertson NM*, Sharma AK*, **Mingling Yang**, Das SK, Siddharthan T, Pollard SL, Rykiel NA, Alupo P, Flores-Flores O, Kirenga B, Chandyo RK, Quaderi SA, Shrestha L, Wise RA, Hurst JR, Checkley W; Global Excellence in COPD outcomes (GECO) Study Investigators. A Multi-country Cohort Study Evaluating the Prevalence, Risk Factors, Lung Function and Clinical Outcomes of Chronic Bronchitis in Low- and Middle-income Countries. *European Respiratory Journal*. 2026 Mar; 67(3):2501435. doi: 10.1183/13993003.01435-2025. PMID: 41381225.
4. Nicolaou L, Reuland CJ, **Mingling Yang**, Williams KN, Hartinger SM, Chiang M, Checkley W. Long-Term Effects of Cooking with Liquefied Petroleum Gas or Biomass on Linear Growth Trajectories from Birth to the Pre-School Years in Puno, Peru. *The Lancet Regional Health – Americas*. 2026 Mar; 55:101382. doi:10.1016/j.lana.2026.101382.
5. Checkley W, Robertson NM, **Mingling Yang**, Nicolaou L, Wise RA, Sharma AK, Chandyo RK, Shrestha LP,

- Das SK, Kirenga B, Alupo P, Gianella G, Siddharthan T, Pollard SL, Quaderi S, Rykiel N, Flores-Flores O, Hurst JR; Global Excellence in COPD outcomes (GECO) investigators. Prevalence of Chronic Obstructive Pulmonary Disease and Population Attributable Risk across Three Resource-Limited Settings: The Global Excellence in COPD Outcomes (GECO) Study. *The Lancet Global Health*. (Accepted)
6. Robertson N, Sharma A, **Mingling Yang**, Das S, Siddharthan T, Pollard S, Rykiel N, Alupo P, Flores-Flores O, Kirenga B, Chandyo R, Quaderi S, Shrestha L, Wise R, Hurst J, Checkley W. Reframing bronchodilator responsiveness using Z-scores: Multi-country evaluation of Z-score-based criteria. *European Respiratory Journal*. (Accepted)
 7. Emetu SA, **Mingling Yang**, Nicolaou L, Kephart JL, Fandiño-Del-Río M, Wise RA, Williams KN, Steenland K, Koehler K, Checkley W; CHAP Trial Investigators. Effects of Cooking with Liquefied Petroleum Gas or Biomass on Lung Function in Adult Peruvian Women: A Randomized Controlled Trial. *American Journal of Respiratory and Critical Care Medicine*. (Under review).

Manuscripts in Preparation:

1. **Mingling Yang**, Laura N. Building a High-Resolution PM2.5 Map for Bhaktapur, Nepal: A Dynamic Spatiotemporal Calibration Model for Low-Cost Sensors. (Preparing for submission)
2. **Mingling Yang**, Luu P, Nicolaou L, Williams K, Checkley W. A Multilevel Functional Data Model for Nocturnal Oxygen Desaturation and Cardiometabolic Outcomes in High-altitude Populations. (Preparing for submission)
3. Williams K*, **Mingling Yang***, Kephart JL, Mallikarachchi M, Fandiño del Río M, Koehler K, Harvey SA, Checkley W. Social Networks and Clean Fuel Use Behaviours in Rural Peru: a Social Network Analysis of the CHAP Trial. (Preparing for submission)
4. Wells M*, Tarazona-Meza C*, **Mingling Yang**, Williams K, Nicolaou L, Steenland NK, Checkley W. Assessing the Impact of Vitamin A and Omega-3 Fatty Acid Deficiencies on Eye Health Among Women Over 40 in the HAPIN Cohort. (Preparing for submission)
5. Luu P, **Mingling Yang**, Checkley W. Cardiometabolic Impact of Nocturnal Hypoxia in High Altitude Sleep Disordered Breathing. (Preparing for submission)
6. Simkovich SM*, Nicolaou L*, **Mingling Yang**, Checkley W. The Effect of a Liquefied Petroleum Gas Stove Intervention on Quality of Life in Guatemala, India, Peru and Rwanda. (Preparing for submission)

Abstracts and Presentations:

1. **Mingling Yang**, N.M. Robertson, A.K. Sharma, R.K. Chandyo, L. Shrestha, S.K. Das, B. Kirenga, P. Alupo, G.E. Gianella, T. Siddharthan, S.L. Pollard, S. Quaderi, N. Rykiel, O. Flores-Flores, J.R. Hurst, R.A. Wise, and W. Checkley. Screening for Chronic Obstructive Pulmonary Disease (COPD) Using the St. George's Respiratory Questionnaire. *Am J Respir Crit Care Med*. 2025;211:A1247. doi: <https://doi.org/10.1164/ajrccm.2025.211.Abstracts.A1247>
2. **Mingling Yang**, N.M. Robertson, A.K. Sharma, R.K. Chandyo, L. Shrestha, S.K. Das, B. Kirenga, P. Alupo, G.E. Gianella, T. Siddharthan, S.L. Pollard, S. Quaderi, N. Rykiel, O. Flores-Flores, J.R. Hurst, R.A. Wise, and W. Checkley. Screening for Chronic Obstructive Pulmonary Disease (COPD) Using the St. George's Respiratory Questionnaire. Presented at: *American Thoracic Society International Conference*; 2025 May 18; San Francisco, CA, USA.
3. Robertson N*, **Mingling Yang**, et al. Chronic Bronchitis Prevalence, Risk Factors, and Outcomes in a Multinational Cohort. *Eur Respir J*. 2025;66:Suppl.69, PA463.; doi: <https://doi.org/10.1183/13993003.congress-2025.PA463>
4. Robertson N*, **Mingling Yang**, et al. Chronic Bronchitis Prevalence, Risk Factors, and Outcomes in a Multinational Cohort. Presented at: *European Respiratory Society International Congress*; 2025 Oct 1; Amsterdam, Netherlands.
5. Wells M, **Mingling Yang**, Nicolaou L, Williams K, Checkley W. Dry Eye Symptoms and Household Air Pollution: Findings from the HAPIN Trial in Peru. Presented at: *Women in Ophthalmology Summer Symposium*; 2025 Aug 12; Amelia Island, FL, USA.
6. Emetu S, **Mingling Yang**, Nicolaou L, Kephart J, Fandiño-del-Río M, Williams K, Steenland NK, Koehler K, Checkley W. Cooking with liquified petroleum gas versus biomass on lung function in adult Peruvian women. Presented at: *CHEST 2026 Annual Meeting*; 2026 Oct 18–21; Phoenix, AZ, USA.

RESEARCH EXPERIENCE

Rollins School of Public Health, Emory University

Data Analyst | Collaborator: Kyle Steenland, Professor of Environmental Health & Epidemiology Jan 2025 – Present

- **Household Air Pollution Intervention Network (HAPIN) Trial** Guatemala, India, Peru, Rwanda
 - Led data management, data visualization, and socioeconomic index (SES) construction using Multiple Correspondence Analysis (MCA) across Guatemala, India, Peru, and Rwanda to quantify household- and community-level disparities.
 - Replicated analyses in R Markdown and cross-validated with original SAS outputs

- Prepared the manuscript draft and collaborated with co-author to address the comments of peer-reviewed article.
- **Output:** Contributed to the article titled "Exposures to Household Pollution From Biomass Cooking and Severe Pneumonia in Infants." **JAMA Network Open.** 2025; 8(10): e2538721.

Bloomberg School of Public Health, Johns Hopkins University

Research Assistant | Advisor: Laura Nicolaou, Professor of Environmental Health & Engineering Dec 2023 – Present

- **Alliance for a Healthier World (AHW) - Ambient Air Pollution Project** Nepal & Peru
 - Led multiple projects for ambient air pollution using ambient monitoring network and E-sampler data over four years ($n \approx 75,000,000$).
 - Developed cross-regional calibration models for co-located PurpleAir low-cost PM_{2.5} sensors against MetOne E-Sampler reference monitors using geostatistical (Kriging) and machine-learning algorithms (Random Forest, XGBoost) across Nepal and Peru; performed in-field sensor co-location, deployment, and reference-grade calibration data collection.
 - Created dynamic spatiotemporal models integrating meteorological covariates to generate hourly PM_{2.5} maps to analyze seasonal variations and urban-rural disparities, improving cross-sensor reliability and enabling effective pollution tracking in low- and middle-income countries (LMICs).
 - **Output:** Manuscript in preparation titled "Building a High-Resolution PM_{2.5} Map for Bhaktapur, Nepal: A Dynamic Spatiotemporal Calibration Model for Low-Cost Sensors."
- **HAPIN - Longitudinal Study of Early Child Growth** Puno, Peru
 - Developed a causal inference framework using directed acyclic graphs (DAGs) to identify a single-pollutant exposure-response model for PM_{2.5}, CO, and early childhood development.
 - Examined associations between air pollution and height-for-age (HAZ) trajectories via longitudinal mixed-effects models, using HAPIN randomized controlled trial with three-year follow-up ($n=683$).
 - Published online supplement by implementing reproducible R Markdown workflows for data cleaning, visualization, modeling, and analytic validation.
 - **Output:** Co-author; Published in **The Lancet Regional Health - Americas** (2025).

School of Medicine, Johns Hopkins University

Research Assistant | Advisor: William Checkley, Professor of Medicine & Epidemiology & Biostatistics Sep 2023 – Present

- **Cardiopulmonary outcomes and Household Air Pollution (CHAP) - Social Network Study** Peru
 - Led the CHAP-Social Network Project and served as first or co-first author on the manuscript.
 - Investigated how social network structure, tie strength, and aspirational ties influence long-term adoption and sustained use of liquefied petroleum gas (LPG) stoves in rural Peruvian communities.
 - Built harmonized behavioral and network databases integrating household, exposure, and relational data for multi-level modeling, using random forest to identify predictors of LPG adoption and diffusion ($n=151$).
 - Developed reproducible R Markdown pipelines and codebooks to standardize analyses across field sites.
 - **Output:** Manuscript in preparation titled "Social Networks and Clean Fuel Use Behaviours in Rural Peru: a Social Network Analysis of the CHAP Trial."
- **HAPIN - Eye and Nutrition Substudy** Puno, Peru
 - Investigated the association between UV exposure, vitamin A/omega-3 deficiency, and ocular diseases (cataract, pterygium) among rural women in Peru ($n=798$).
 - Designed integrated data structures combining ophthalmologic exams, nutritional biomarkers, and environmental exposure records.
 - Visualized regional variation in ocular disease prevalence using GIS-based spatial mapping.
 - **Output:** Poster presentation at **omen in Ophthalmology Summer Symposium 2025**.
- **Global Respiratory Health Research (Chiesi)** Nepal & Peru
 - Designed and supported the data collection of respiratory system diseases to expand spirometry/oscillometry screening among brick kiln workers across Nepal and Peru.
 - Designed and maintained REDCap databases to integrate multi-level data, including household, facility capacity, and individual clinical forms.
 - Led data management, quality control, and visualization processes; collaborated with local works on REDCap workflows and data reporting.
 - **Output:** Database infrastructure adopted for Chiesi Global Lung Health Program (2024–2026).
- **Global Excellence in COPD Outcomes Study (GECOPD) - Chronic Bronchitis Prevalence and Risk Factor Modeling** Nepal & Peru & Uganda

- Investigated the modifiable risk factors of COPD and chronic bronchitis using longitudinal data from three counties among diverse LMIC populations ($n = 9,664$).
- Conducted logistic regression and machine-learning analyses (Random Forest) using harmonized survey and spirometry data; validated models through cross-validation and sensitivity testing.
- Contributed evidence positioning chronic bronchitis as an early clinical marker of COPD progression.
- Emphasized the crucial effect of socioeconomic disparities in the prevention and control of COPD.
- **Output:** Co-author; Published in **European Respiratory Journal**; Poster presentation at **European Respiratory Society Congress 2025**.
- **GEC - Population Attributable and Impact Fractions of COPD Risk Factors** Nepal & Peru & Uganda
 - Quantified the contribution of behavioral, environmental, and clinical risk factors to COPD burden using Population Attributable Fractions (PAFs) and Potential Impact Fractions (PIFs) ($n = 10,709$).
 - Built integrated databases and a suite of custom R functions to standardize variable mapping, automate multi-level data linkage, and implement reproducible meta-analytic workflows across sites.
 - Modeled the combined effect of a simultaneous 25% reduction in multiple risk factors using PIFs, incorporating a causal pathway approach to account for risk factor interactions.
 - **Output:** Co-author; Published in **The Lancet Global Health** (2025).
- **GEC - Optimizing COPD Screening Using the SGRQ** Nepal & Peru & Uganda
 - Evaluated the performance of the St. George’s Respiratory Questionnaire (SGRQ) as a population screening tool for COPD compared with GOLD-defined diagnosis.
 - Generated simulations for ROC and AUC analysis to evaluate the performance of SGRQ and PEF as screening tools, achieving 90% sensitivity based on Youden Index using R.
 - Built optimized algorithm for the PDF text extraction technique to address 30% missing data in a large, heterogeneous cohort of adults from Nepal, Peru, and Uganda, ensuring credible and rigorous statistical inferences. ($n=10,709$)
 - Demonstrated that $SGRQ \geq 25$ predicted spirometry-defined COPD with 91% sensitivity and 98% negative predictive value.
 - Presented the poster at the American Thoracic Society International Conference 2025.
 - **Output:** Co-author; Published in **American Journal of Respiratory and Critical Care Medicine** (2025).
- **HAPIN - Neurodevelopmental Outcomes in Infants** Puno, Peru
 - Conducted E-R analyses linking prenatal and postnatal $PM_{2.5}$ and CO to early childhood development, including Bayley-III cognitive, motor, and language scores, using HAPIN trial ($n \approx 301$).
 - Implemented modified intention-to-treat models to preserve causal inference and applied multiple imputation for missing neurodevelopmental assessments.
 - Built integrated databases and a suite of custom R functions to standardize variable mapping, automate multi-level data linkage, and implement reproducible meta-analytic workflows across diverse study sites, improving scalability and analytic reproducibility.
 - **Output:** Co-author; Published by **Environmental Health Perspectives** (2025).

School of Medicine, Johns Hopkins University

Research Assistant | Advisor: Prof. Luu Pham, Professor of Medicine

Sep 2024 – Present

- **Nocturnal Hypoxia and Cardiometabolic Risk Study (NHCRS)** Puno, Peru
 - Characterize nocturnal oxygen desaturation patterns and their associations with cardiometabolic biomarkers among women living at high altitude ($n = 120$).
 - Applied functional data analysis (fPCA) to continuous SpO_2 curves to extract latent hypoxia phenotypes and modeled their relationship with blood pressure, glucose, and lipid markers using hierarchical regression.
 - Built and managed the full data pipeline for field-collected SpO_2 and posture adherence signals—developed MATLAB and R scripts for signal extraction and quality scoring.
 - Preliminary results suggest distinct desaturation trajectory clusters linked to higher systolic BP and fasting glucose. Simulation studies underway to quantify model uncertainty and validate fPCA-based inference.
 - **Output:** Manuscript in preparation titled “A Multilevel Functional Data Model for Nocturnal Hypoxia and Cardiometabolic Outcomes in High-Altitude Populations.”

WORKING EXPERIENCE

- **Johns Hopkins University**

Baltimore, MD

Senior Research Data Analyst, Prof. William Checkley: Biostatistics & Epidemiology

Sep 2023 – Present

- **Global Environmental Health Analytics:** Served as lead data analyst across 9 multinational studies. Managed, harmonized, and analyzed over 100 million data points spanning environmental exposure, clinical health, sensor data, questionnaire data, biomarker data, and social datasets.
- **Pipeline Development:** Designed reproducible R/SQL workflows and internal R packages to automate data integration, cleaning, and calibration for large multi-country trials, improving reproducibility, documentation, and QC efficiency by over 60%.
- **Statistical Methods & Modeling:** Applied advanced methods including linear mixed models, functional data analysis, Bayesian hierarchical modeling, and causal DAGs for multi-level environmental and health data.
- **Data Architecture & REDCap Integration:** Built multi-tier REDCap systems linking household, individual, and facility-level data for Chiesi Foundation projects; trained local researchers on REDCap quality control and follow-up tracking.
- **Visualization & Automation:** Developed reproducible R workflows and internal packages to automate data cleaning, visualization, and reporting across multi-country studies. Produced statistical reports, exposure maps, and interactive dashboards (ggplot2, leaflet, Shiny) supporting WHO-aligned dissemination and reproducibility.
- **Mentorship & Collaboration:** Coordinated analysis meetings between JHU, Emory, UC-Berkeley, Harvard, and local field teams; supervised junior data analysts and contributed to manuscript methodology sections.
- **Publications:** Published 5 papers in peer-reviewed journals.

- **Qiangyuan Sporting Goods Co.**

Zhuhai, China

Data Scientist, Full time

Jan 2021 - Sep 2021

- **Data Infrastructure & Quality Control:** Improved data integrity across three production divisions by mapping process workflows, conducting staff interviews, and implementing Python- and R-based validation scripts to identify and correct inconsistencies in large-scale manufacturing datasets.
- **Predictive Analytics & Forecasting:** Applied statistical modeling and machine-learning algorithms to uncover material-inventory dynamics and build demand-forecasting models, reducing excess inventory and storage costs by approximately 20%.
- **Database Engineering:** Designed and administered SQL databases integrating procurement, production, and sales data—establishing the company’s first centralized analytics platform to support real-time data querying and decision-making.
- **Process Automation:** Streamlined manual Excel workflows into automated Python/R pipelines, generating monthly financial and procurement summaries and increasing reporting efficiency by over 80%.

- **Shandong University, Shandong Big Data Research Association**

Jinan, China

Research Assistant, Full-time, Prof. Yufeng Shi: Statistics & Finance

Jan 2021 - May 2021

- **Research Coordination:** Led a 12-member team through project presentations and methodological reviews, managing timelines, revision workflows, and collaborative analysis across multiple research groups.
- **Quantitative Research & Publication:** Assisted in the preparation of the second edition of *Quantitative Investment*, conducting data validation, literature synthesis, and empirical case updates on portfolio optimization and statistical trading models.

TEACHING EXPERIENCE

- **Department of Applied Mathematics and Statistics, Johns Hopkins University**

Baltimore, MD

Teaching Assistant: Introduction to Data Science

Jan 2022 - May 2022

- Assisted in lab instruction, assignment design, and project grading.
- Supported students in statistical learning methods (regression, SVMs, PCA) and Python implementation.

- **Department of Applied Mathematics and Statistics, Johns Hopkins University**

Baltimore, MD

Teaching Assistant: Computing for Applied Mathematics

Sep 2022 - Dec 2022

- Taught weekly sessions, guided weekly office hours, and designed weekly quizzes for ~30 graduate students covering Python programming, numerical methods, and data manipulation for applied mathematics.
- Provided support in debugging, optimizing code, and interpreting computational results while grading weekly assignments, quizzes, and final projects with detailed feedback on both correctness and code style.

TECHNICAL SKILLS

- Language: GRE V164 Q170
- Programming & Statistical Computing: R, Python, MATLAB, SAS, SQL, JAVA, RMarkdown, LaTeX, Git, AWS, REDCap, Tableau
- Geospatial Analysis: ArcGIS, R sf/terra/raster packages, custom shapefile processing, spatial interpolation (Kriging), GIS-based exposure surface generation
- Environmental Monitoring & Field Instrumentation: PurpleAir low-cost PM2.5 sensor co-location, deployment, and reference-grade calibration against MetOne E-Sampler monitors; ambient air-quality network design and quality control for LMIC field campaigns

PROFESSIONAL ACTIVITIES AND INVITED PARTICIPATION

- NIH Fogarty Global Health Program for Fellows and Scholars / Launching Future Leaders in Global Health Research Training Program (LAUNCH) — Orientation and Training Bethesda, MD, July 7–11, 2025
 - Attended the NIH Fogarty International Center–convened LAUNCH Orientation as a member of the Johns Hopkins research group (D43TW009340 UJMT consortium / D43TW011502 JHU–Peru Respiratory Training consortium).
 - Observed the 2025 LAUNCH Orientation – PI Council Meeting (Hyatt Regency Bethesda, July 8, 2025).
- Professional Society Memberships
 - American Thoracic Society (ATS) 2025–present
 - Minorities in Agriculture, Natural Resources, and Related Sciences (MANRRS) 2025–present
 - International Society for Environmental Epidemiology (ISEE) 2026–present

REFERENCE

- Dr. William Checkley, Dr. Laura Nicolaou, Dr. Kyle Steenland